More Foresight, Induced Collapse: Response Horizons in Self-Consuming Model Optimization

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Abstract

When a model's own outputs become its future training data, how much of that feedback should the optimizer anticipate? The standard answer is to anticipate all of it, differentiating through the distribution the model eventually induces. This paper shows that answer is wrong in a specific and instructive way: anticipating the full response is exactly equivalent to optimizing an objective that rewards the model for reducing the diversity of its own output, and the resulting collapse threshold in the synthetic-data fraction therefore falls as foresight deepens. Because the result holds with exact gradients and no estimation noise, the failure lies in the objective rather than in the estimate. A recursive MNIST autoencoder sharpens the picture: deeper foresight steadily improves class balance while cutting generated variance to 27.5% of the real value, so a monitor that counts modes reports a healthy model while diversity collapses inside each one. The full response is also fragile to estimate when the environment reacts slowly, and a short horizon is then more accurate under a matched budget. Response depth is therefore best understood as a modeling choice about how much of its own influence a learner should internalize, not as an approximation error to be minimized.

Keywords: self-consuming generative models, performative prediction, model collapse, response horizon, implicit differentiation, bifurcation, synthetic data

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1 Introduction

Standard supervised learning assumes a data distribution that does not depend on the model. Many deployed systems violate this. A credit model changes how applicants manage credit lines; a recommender changes what users want to watch; a generative model fills a corpus that the next generative model trains on. In each case the data faced tomorrow depends on the model shipped today.

Once this is acknowledged, a design question appears that has no counterpart in ordinary training. The learner can account for its own influence to a greater or lesser degree. It can ignore the feedback entirely and treat the environment as frozen. It can account for the full eventual reaction, asking where the environment finally settles once it has finished responding to a fixed model. Or it can account for the reaction over some bounded number of steps and stop there. The literature has converged on the second option as the correct target, and the purpose of this paper is to argue that it is not always the right one.

Let x denote an environment or population state and θ the learner's parameters. The environment responds through a map $x_{k+1} = \Phi(x_k, \theta_k)$. When a frozen-parameter fixed point $x^*(\theta)$ exists, the total derivative of the long-run objective $\bar{F}(\theta) = F(x^*(\theta), \theta)$ contains the fixed-point sensitivity

$$\nabla \bar{F} = F_\theta + \Phi_\theta^\top (I - \Phi_x)^{-\top} F_x. \tag{1}$$

This is the *stationary response*, and it is prior art. It is the sensitivity underlying PerfGD and Stateful PerfGD (Izzo et al., 2021, 2022), and it coincides with the hypergradients of implicit differentiation and bilevel optimization (Grazzi et al., 2020; Blondel et al., 2022; Grazzi et al., 2024). Both literatures treat it as the object one would use if it were free, with truncation justified by memory or compute (Shaban et al., 2019; Metz et al., 2019).

The alternative studied here is to propagate the response for a bounded number of steps K and truncate, through the recursion

$$v_0 = 0, \qquad v_{j+1} = F_x + \Phi_x^\top v_j, \qquad g_K = F_\theta + \Phi_\theta^\top v_K. \tag{2}$$

At $K = 0$ this is ordinary frozen-environment training; as K grows it approaches (1) geometrically under a contraction assumption. The number K is called the *response horizon* throughout. Rather than asking how well a finite horizon approximates the stationary endpoint, this paper asks whether that endpoint is the horizon a learner should be aiming at in the first place.

Two separate answers emerge, and they matter because they fail for different reasons and admit different remedies.

The first concerns the objective being optimized, and it is the paper's central result. In a self-consuming generative loop, where a model's own samples form part of its next training set, the depth-K update turns out to be exactly the gradient of a modified objective. That objective is the ordinary real-data loss plus a term proportional to the entropy of the model's own generative distribution, with the proportionality constant growing from zero at $K = 0$ to its maximum at the stationary endpoint. Minimizing it therefore rewards the model for becoming less varied, and rewards it most strongly when the model is most thorough about anticipating its own feedback. The consequence is a closed-form threshold in the synthetic-data fraction at which the balanced state loses stability, and that threshold falls as foresight deepens: full internalization collapses at half the synthetic fraction that naive training tolerates. Nothing here is approximated and no quantity is estimated from noisy samples. The optimizer is computing the correct gradient of a badly posed

problem, which is why this is described as a specification failure rather than a numerical one, and why better data does not remedy it.

The second answer concerns estimation. The stationary correction requires inverting a matrix whose gain is the norm of the resolvent $(I - \Phi_x)^{-1}$. Under the standard linear-relaxation response this gain is exactly the reciprocal of the environment's reaction rate, and it diverges as that environment slows. In the slowest benchmark considered below the gain is 111.68, so an error in the estimated one-step response is multiplied roughly a hundredfold before it reaches the update direction. A slowly reacting environment is precisely the case where the observable one-step effect is small and almost all of the eventual response must be inferred by that multiplication, so the method is least reliable exactly where it is most needed. A fixed finite horizon accepts a known and bounded truncation bias in exchange for never resolving the near-unit pole. Unlike the first failure, this one is an estimation problem and does recede with better data.

The empirical centerpiece is a recursive MNIST autoencoder, reported in Section 6.1. It shows the two diagnostics one might naturally watch moving in opposite directions as foresight deepens. Class balance steadily improves, while the variance of the generated images steadily falls to roughly a quarter of the real value. Both digit classes survive the whole way; each one collapses internally onto a single prototype. An operator checking whether the model still covers its classes would therefore see a healthy and improving answer throughout a diversity collapse. Mode coverage is not a diversity diagnostic, and this is argued to be the most immediately actionable finding in the paper.

Three further results support these arguments. Section 4 characterizes exactly when a finite horizon degenerates into a rescaling of the stationary correction, a non-generic condition on the response map, and constructs feedback outside that class. Section 5 separates truncation bias from estimation error in a single bound, derives an explicit preferred depth, and supplies a finite-sample uncertainty radius for the rolling regression Stateful PerfGD actually uses.

1.1 Related work

Performative prediction. The setting was introduced by Perdomo et al. (2020), who studied repeated risk minimization as a route to a performatively stable point. PerfGD (Izzo et al., 2021) was the first algorithm shown to converge to the performative optimum rather than merely a stable point. Stateful PerfGD (Izzo et al., 2022) is closest to the present setting: the population reacts gradually rather than instantaneously, and its stationary sensitivity is estimated from rolling response observations along a single trajectory. Subsequent work covers stateful populations (Brown et al., 2022), neural predictors (Mofakhami et al., 2023), Markovian data (Liu et al., 2024), plug-in optimization (Lin and Zrnic, 2024), and performative control (Cai et al., 2024).

Truncated hypergradients. This is the closest prior art, and the boundary deserves care. Truncating a fixed-point differentiation is standard in bilevel optimization and meta-learning. Shaban et al. (2019) analyze truncated back-propagation and show that truncated hypergradients remain sufficient descent directions; Grazzi et al. (2020) give iteration-complexity results; Metz et al. (2019) document pathologies of unrolled optimization at long horizons. In all of that work, truncation is motivated by memory and compute, and the exact hypergradient is the thing one would use if it cost nothing.

The present claim differs in three respects. First, truncation is statistically preferable in an identified regime—a slowly reacting environment with noisy estimation—for reasons that have nothing to do with compute, because the stationary object requires resolving a pole near unity.

Second, the horizon enters the stability boundary of the coupled learner–environment system, so it selects which attractor the joint dynamics reach rather than only how accurate a single direction is. Third, in self-consuming systems the exact hypergradient optimizes a distorted objective, which makes it not merely expensive but undesirable. None of these three appears in the truncation literature.

Self-consuming training. Recursive use of model-generated data degrades quality or diversity (Alemohammad et al., 2024; Shumailov et al., 2024), with subsequent theory on collapse rates (Fu et al., 2024; Suresh et al., 2025; Dohmatob et al., 2025). A central practical distinction is between *replacing* real data with synthetic data and *accumulating* both: Gerstgrasser et al. (2024) and Kazdan et al. (2025) show that under accumulation the test error admits a bound independent of the number of iterations. Curation and filtering can also contain the effect (Gillman et al., 2024; Wei and Zhang, 2025; Cai et al., 2026). Whether collapse is a practical threat remains genuinely disputed and this paper takes no side; the results here concern which objective a learner optimizes under feedback, a separate matter from whether any particular corpus is at risk.

Dynamics of learning. Transitions from stability to instability and chaos in multi-agent performative prediction are established by Piliouras and Yu (2023), and large-learning-rate spectral effects beyond small-step optimization by Wang et al. (2025). Estimation from a single dependent trajectory has its own literature in system identification and self-normalized least squares (Simchowitz et al., 2018; Abbasi-Yadkori et al., 2011), while noisy regressors produce errors-in-variables effects that require different tools from independent-design least squares (Rudelson and Zhou, 2017).

Novelty boundary. Several ingredients here are individually standard and it is worth saying so plainly: the stationary derivative (1), its truncation by a Neumann series, model collapse in recursive training, and the existence of transitions in self-consuming training, the last of which Fu et al. (2024) and Khelifa et al. (2026) already establish in different quantities. What is claimed new is the identification of the response horizon as the coefficient tempering a learner's self-entropy reward, the threshold and variance laws that follow, and the empirical finding that coverage and within-mode diversity move in opposite directions as that horizon deepens.

2 Setup

Let Φ denote the one-step response map, with partial Jacobians Φ_x and Φ_θ, and let $F(x, \theta)$ be the training potential. Write $S_\infty = (I - \Phi_x)^{-1}\Phi_\theta$ for the stationary state sensitivity and $S_K = \sum_{j=0}^{K-1} \Phi_x^j \Phi_\theta$ for its depth-K truncation. Under a linear relaxation of rate δ it is convenient to write $\alpha_K := 1-(1-\delta)^K$, a quantity rising from zero at $K = 0$ to one in the limit; it recurs throughout as the natural measure of how much of the eventual response a given horizon captures. The depth-K update direction is g_K, defined in (2).

The structural feature that makes this setting different from ordinary optimization is worth stating explicitly, because several later results depend on it. The two blocks of the system move for different reasons. The learner descends an objective. The environment does not descend anything; it follows its own law and optimizes nothing at all. There is consequently no single function that both blocks are decreasing, which is what permits the multiplicity of equilibria and the loss of stability studied below, and what makes the joint spectrum rather than the learner's Hessian the object to examine.

Assumption 1 (Contractive response)**.** *For each θ in an open set, the map $x \mapsto \Phi(x, \theta)$ is continuously differentiable and satisfies $\|\Phi_x\| \le q < 1$.*

Under Assumption 1 the fixed point is unique, (1) holds, and a Neumann expansion bounds the truncation error geometrically, by $\|\Phi_\theta\|\, q^K (1-q)^{-1} \|F_x\|$. Truncation bias therefore decays at a known rate, which makes it a controllable cost.

3 Objective distortion: the entropy-tempering theorem

3.1 A self-consuming two-mode generator

The simplest setting that exhibits the effect is a generator with two well-separated modes that trains partly on its own samples. Let the real distribution place equal mass on each mode, and let the generator learn only the mixture weight $p(z) = (1+e^{-z})^{-1}$. Up to constants the relevant part of the negative log likelihood is the Bernoulli cross entropy $F(z; x) = -x \log p(z) - (1-x) \log[1-p(z)]$, where x is the frequency of the positive mode in the current training pool. The pool mixes real and synthetic samples and relaxes toward its target at rate δ:

$$x_{n+1} = (1-\delta) x_n + \delta \Big[\tfrac{1-\beta}{2} + \beta p(z_n) \Big], \qquad 0 < \delta \le 1, \tag{3}$$

with β the fraction of the pool that is synthetic. The partial derivatives follow immediately, giving $S_K = \beta \alpha_K p(1-p)$ and an update $z_{n+1} = z_n - \eta g_K$ with $g_K = p - x - \beta \alpha_K p(1-p) z$. The balanced generator, which places equal weight on both modes, is always a stationary point. The question is when it stops being a stable one.

Theorem 2 (Response-depth threshold for symmetry breaking)**.** *Let $0 < \delta \le 1$ and $K \in \{0, 1, \dots, \infty\}$. The balanced point $(x, z) = (1/2, 0)$ of (3) has a $+1$ multiplier at*

$$\boxed{\beta_c(K) = \frac{1}{1 + \alpha_K}} \tag{4}$$

For $0 < \beta < \beta_c(K)$ it is locally asymptotically stable whenever $0 < \eta < 8(2-\delta)[(2-\delta)(1-\beta\alpha_K) + \delta\beta]^{-1}$. At $\beta = \beta_c(K)$ the critical multiplier crosses $+1$ transversely and the equilibrium equation undergoes a supercritical symmetry-breaking pitchfork; for $\mu = \beta - \beta_c(K) > 0$ small and $K \ge 1$, the emerging branches satisfy $z_\pm(\beta) = \pm\sqrt{6(1+\alpha_K)^2 \mu / \alpha_K} + O(\mu^{3/2})$.

Proof. At the balanced point the joint Jacobian is $J_K = \left(\begin{smallmatrix} 1-\delta & \delta\beta/4 \\ \eta & 1-\eta(1-\beta\alpha_K)/4 \end{smallmatrix}\right)$. The Jury condition $1 - T + D = \frac{\delta\eta}{4}[1 - \beta(1+\alpha_K)]$ vanishes exactly at (4). Writing $P(\lambda, \beta) = \lambda^2 - T\lambda + D$, differentiation gives $\partial_\beta P(1, \beta_c) = -\frac{\delta\eta}{4}(1+\alpha_K) \neq 0$ and $\partial_\lambda P(1, \beta_c) = \delta + \eta/[4(1+\alpha_K)] > 0$, so the critical multiplier has strictly positive derivative in β and the crossing is transverse. Expanding $p(z) - \frac{1}{2} = \frac{z}{4} - \frac{z^3}{48} + O(z^5)$ and $p(1-p) = \frac{1}{4} - \frac{z^2}{16} + O(z^4)$ in the stationary equation produces a cubic coefficient equal to $\alpha_K / [24(1+\alpha_K)]$, which is positive, giving the supercritical orientation. □

The threshold is monotone in depth, running from $\beta_c(0) = 1$ down to $\beta_c(\infty) = 1/2$. A learner that ignores its own feedback never breaks symmetry over the admissible range; one that fully internalizes it breaks at half synthetic data. Deeper correction internalizes more of the self-generated pool and therefore destabilizes the balanced state at a smaller synthetic fraction. Table 1 lists exact values at the relaxation rate used in the experiments.

	direct (K=0)	K=1	K=2	K=5	full response
β_c (weight pitchfork)	1.000	0.870	0.783	0.643	0.500
$\beta_{0.75}$ (25% variance compression)	1.000	0.690	0.546	0.375	0.250

Table 1: Exact thresholds from Theorem 2 and Proposition 4 at $\delta = .15$. Deeper response produces both earlier variance compression and earlier symmetry breaking.

3.2 The surrogate objective

Theorem 2 establishes the effect. The result below identifies its cause, and is the central claim of the paper.

Allow the mixture components themselves to be learned rather than held fixed, so that the generator controls the mode weight, both component locations, and both component variances. Assume the modes are separated enough that component identities can be tracked, an idealization whose likelihood cost is exponentially small in the separation. Summarize the training pool by its sufficient statistics, which relax toward a real-synthetic mixture at the same rate as before.

The key observation is that the expected direct score under the model's own distribution vanishes, as it must. What survives in the response term is therefore not a fitting term at all, but the gradient of the entropy of the model's own generative distribution,

$$H_{\text{lab}}(\theta) = -p \log p - (1-p)\log(1-p) + \tfrac{p}{2}\log(2\pi e v_-) + \tfrac{1-p}{2}\log(2\pi e v_+), \tag{5}$$

carrying a coefficient that depends on the response horizon.

Proposition 3 (Exact surrogate objective)**.** *Let $C_0(\theta)$ denote the real-data labelled cross entropy. At the stationary pool associated with parameter θ, the depth-K response-aware gradient satisfies*

$$g_K^*(\theta) = (1-\beta)\nabla C_0(\theta) + \beta\alpha_K \nabla H_{\text{lab}}(\theta) = \nabla J_K(\theta), \qquad \boxed{J_K(\theta) = (1-\beta)C_0(\theta) + \beta\alpha_K H_{\text{lab}}(\theta).} \tag{6}$$

Proof. At the stationary pool the direct score contributed by the synthetic fraction vanishes, since the expected gradient of the negative log likelihood under the model's own distribution is zero. The direct part is therefore $(1-\beta)\nabla C_0$. The response term contributes $\beta\alpha_K \nabla H_{\text{lab}}$, and adding the two gives (6). □

The interpretation is the point. The response horizon exactly tempers how strongly the optimizer internalizes the entropy of its own output. At $K = 0$ the self-entropy term is absent entirely. At the stationary endpoint it carries its full weight β. Since the learner is minimizing J_K, and since entropy enters with a positive coefficient, the learner is being paid to make its own output less varied, and paid most at full foresight.

It follows that full stationary-response optimization is faithfully optimizing (6). Nothing is approximated badly and no quantity is estimated from noisy samples. The endogenous objective itself increasingly rewards low-entropy self-generated data as the horizon deepens. Collapse in this model is not a numerical failure but the correct solution to a distorted problem, which is the sense in which it is a specification phenomenon and not an estimation one. This distinction matters practically: an estimation failure recedes with better data or a longer run, whereas this one does not.

Proposition 4 (Variance compression on the symmetric branch)**.** *On the reflection-symmetric stationary branch, where the mode weight is one half and the two component variances are equal, the learned variance satisfies*

$$\frac{v_K^*(\beta)}{v_0} = \frac{1-\beta}{1-\beta+\alpha_K\beta}. \tag{7}$$

The threshold (4) *is unchanged, and the learned variance has fallen to exactly half the real variance at the moment the weight pitchfork occurs, for every* $K \geq 1$*. The branch reaches a variance fraction* τ *at* $\beta_\tau(K) = (1-\tau)/[(1-\tau)+\tau\alpha_K]$*.*

Proof. On the symmetric branch, the direct derivative of the expected labelled negative log likelihood with respect to the log variance is $\frac{1}{4}(1-\beta)(1-v_0/v)$, while by (5) the response term contributes $\alpha_K\beta/4$. Setting the sum to zero and solving gives (7). The entropy is translation invariant, so the mean equations are unaffected and the component locations remain at their true values. When the two variances coincide, the entropy term in the mode-weight derivative reduces to the Bernoulli case already treated, so Theorem 2 applies unchanged. □

Equation (7) is worth reading alongside the threshold. Variance compression begins immediately and proceeds smoothly, whereas symmetry breaking waits for a critical value. A learner can therefore be losing a great deal of within-mode diversity while its mode weights remain perfectly balanced, which anticipates the empirical finding of Section 6.1.

Remark 5 (This is a pitchfork, not hysteresis)*.* The collapse literature often reaches for the language of irreversibility (Shumailov et al., 2024), so it is worth being precise about what the present transition is and is not. It is supercritical: the symmetry-broken branches emerge continuously with square-root amplitude, and lowering the synthetic fraction back below the threshold restores the balanced state. There is no gap between the forward and return thresholds and no path dependence. Genuine hysteresis—a subcritical branch, a fold, and distinct forward and return thresholds—would require structure this model does not contain, at minimum an additional slow endogenous state that fails to relax when the synthetic fraction is reduced. Establishing or ruling out such a state is the most important extension of this work.

4 Beyond scalar tempering: when depth changes direction

The models of Section 3 relax linearly, which means the depth-K sensitivity is a scalar multiple of the stationary one. Depth changes the strength of the correction but not its direction. This is what makes the closed-form transition results available, but it invites an objection: if the horizon is only a number between zero and one, the horizon language is decoration and a continuous coefficient would serve just as well. The objection is answered in two steps, by characterizing exactly when the degeneracy holds and then constructing feedback outside that class.

Proposition 6 (Scalar-tempering characterization)**.** *Let* $I - \Phi_x$ *be invertible. The following are equivalent.* (i) *There exist scalars* α_K *with* $S_K = \alpha_K S_\infty$ *for all* $K \geq 0$*.* (ii) $\Phi_x\Phi_\theta = \lambda\Phi_\theta$ *for some scalar* λ*, that is, the column space of* Φ_θ *lies inside an eigenspace of* Φ_x*. In that case* $\lambda \neq 1$ *and* $\alpha_K = 1-\lambda^K$*. If instead the column space of* Φ_θ *meets at least two distinct eigenspaces of* Φ_x*, then* S_1 *and* S_2 *are linearly independent and the finite-depth corrections span a Krylov subspace whose dimension grows with* K*.*

Proof. Suppose (ii). Then $\Phi_x^j \Phi_\theta = \lambda^j \Phi_\theta$, so that $S_K = \frac{1-\lambda^K}{1-\lambda}\Phi_\theta$ and $S_\infty = \frac{1}{1-\lambda}\Phi_\theta$, and invertibility forces $\lambda \neq 1$. Conversely, suppose (i). Both $S_1 = \Phi_\theta$ and $S_2 = \Phi_\theta + \Phi_x\Phi_\theta$ are scalar multiples of S_∞, hence of one another, which forces $\Phi_x\Phi_\theta = c\Phi_\theta$ for some scalar c. The Krylov statement is the standard fact that the iterates $\Phi_x^j\Phi_\theta$ remain independent until the minimal annihilating polynomial on the column space is reached. □

This turns an apparent limitation into a structural statement. Scalar tempering is exactly the eigenspace-alignment case, and that case is non-generic. Every recursive pool that relaxes uniformly falls into it, which is precisely why Section 3 admits closed forms. Any response that depends on the current state—curation, filtering, or representation-dependent sample selection—generically leaves it.

A state-dependent family. To exhibit the non-degenerate case concretely, let the synthetic summary be curated in a way that depends on the current pool as well as on the generator:

$$\mathcal{C}(x,\theta) = y(\theta) + \rho\tanh\big(A(x-r) + B(y(\theta)-r)\big), \quad \Phi(x,\theta) = (1-\delta)x + \delta[(1-\beta)r + \beta\mathcal{C}(x,\theta)], \tag{8}$$

where r is the real target summary and $y(\theta)$ the generator's. Differentiating shows that Φ_x acquires a term proportional to A, so unless A is a multiple of the identity the response Jacobian is neither scalar nor constant along the trajectory, and by Proposition 6 the degeneracy fails. The departure can be measured by the angle between the correction a finite horizon produces and the correction the stationary endpoint produces; scalar tempering would force that angle to zero. On the instance studied below, the cosine of that angle falls from .983 at $\beta = .60$ to .754 at $\beta = .85$, .516 at $\beta = .90$, and .118 at $\beta = .95$. Stronger curation rotates the finite-horizon correction away from the stationary direction rather than merely shrinking it.

5 Statistical fragility of the stationary response

Truncation also changes how estimation error propagates into the update, and this matters most precisely when the environment reacts slowly.

Proposition 7 (Stationary inverse versus finite depth)**.** *Let D_1, D_2 be the true short-run response derivatives and let $\widehat{D}_i = D_i + E_i$ be estimates. With $S_\infty = (I - D_2)^{-1}D_1$,*

$$\left\|\widehat{S}_\infty - S_\infty\right\| \le \left\|(I-\widehat{D}_2)^{-1}\right\| \left(\|E_1\| + \|E_2\|\,\|S_\infty\|\right), \tag{9}$$

while if $\|D_2\|$ and $\left\|\widehat{D}_2\right\|$ are both at most $\bar{q}$ then $\left\|\widehat{S}_K - S_K\right\| \le \left(\sum_{j<K}\bar{q}^{\,j}\right)\|E_1\| + \|D_1\|\,\|E_2\|\sum_{j=1}^{K-1} j\bar{q}^{\,j-1}$. For fixed K the finite-depth bound stays finite as $\bar{q}$ approaches one, whereas the stationary bound is controlled by a resolvent norm that diverges.

Proof. For the stationary sensitivity, multiply the error by $I - \widehat{D}_2$ and use $(I - D_2)S_\infty = D_1$. For finite depth, the telescoping identity $\widehat{D}_2^{\,j} - D_2^j = \sum_{\ell<j}\widehat{D}_2^{\,\ell}E_2D_2^{\,j-1-\ell}$ gives the bound $\left\|\widehat{D}_2^{\,j} - D_2^j\right\| \le j\bar{q}^{\,j-1}\|E_2\|$, and summing produces the claim. □

Remark 8 (The slow-response regime). The practical content of Proposition 7 is clearest in the linear-relaxation case. There the state Jacobian is $(1-\delta)I$ and the exact stationary gain is $1/\delta$. The stationary sensitivity is unbiased when the short-run derivatives are known exactly, but a noisy estimate of them is magnified by that factor, which grows without bound as the environment

slows. This is the unfortunate case: a slow response means the one-step effect visible in the data is small, so nearly all of the eventual response must be inferred by multiplying it up, and the error is multiplied along with the signal. The method is least trustworthy exactly where it is most necessary.

Theorem 9 (Stochastic response-depth error and oracle horizon)**.** *Let $\|D_2\| \le q < 1$, and let estimates from n observations satisfy $\mathbb{P}(\|E_i\| > \varepsilon_i(n,\delta_0)) \le \delta_0/2$ with $\varepsilon_i(n,\delta_0) = \sigma_i\sqrt{2\log(4/\delta_0)/n}$. Set $\bar{q}_n := q + \varepsilon_2$, $A_K(r) := \frac{1-r^K}{1-r}$, $B_K(r) := \frac{1-Kr^{K-1}+(K-1)r^K}{(1-r)^2}$, and $C_0 := \|D_1\|/(1-q)$. Then with probability at least $1-\delta_0$, simultaneously for every $K \ge 1$,*

$$\left\|\widehat{S}_K - S_\infty\right\| \le C_0 q^K + A_K(\bar{q}_n)\varepsilon_1 + \|D_1\|\, B_K(\bar{q}_n)\varepsilon_2. \tag{10}$$

The oracle horizon minimizes the right-hand side over any finite candidate set.

Proof. Decompose the error into a truncation part and an estimation part. Since $\|D_2\| \le q < 1$, the deterministic tail satisfies $\|S_K - S_\infty\| \le \sum_{j\ge K} q^j \|D_1\| = C_0 q^K$. A union bound gives $\|E_i\| \le \varepsilon_i$ for both i with probability at least $1-\delta_0$; on that event $\left\|\widehat{D}_2\right\| \le \bar{q}_n$, and Proposition 7 applies for finite K even when the estimated norm exceeds one. □

The two terms on the right of (10) move in opposite directions in K, which gives the bound its bias-variance reading. The first decays geometrically as the horizon lengthens; the second accumulates, and accumulates faster when the response is slow, because short-run derivative errors are propagated repeatedly. More data supports a deeper horizon, and noisier Jacobian estimates call for a shallower one. Corollary 10 makes the resulting optimum explicit in the regime that matters.

Corollary 10 (Preferred depth in the slow regime)**.** *Let $\lambda = -\log q$. In a slow-response regime with a candidate horizon below the mixing time, the root-mean-square version of (10) is approximated by $U_n(K) \approx C_0 e^{-\lambda K} + n^{-1/2}(\sigma_1 K + \frac{1}{2}\|D_1\|\,\sigma_2 K^2)$. If direct-response noise dominates, the minimizer is approximately $\lambda^{-1}[\log(C_0\lambda\sqrt{n}/\sigma_1)]_+$; if state-Jacobian noise dominates, it is approximately $\lambda^{-1}W(C_0\lambda^2\sqrt{n}/(\|D_1\|\,\sigma_2))$, with W the principal Lambert function.*

A radius for the estimator actually used. Theorem 9 requires valid uncertainty radii for the two short-run derivatives but says nothing about where they come from. Supplying them for the estimator Stateful PerfGD uses takes some care. That estimator fits a local Jacobian from a short segment of a single trajectory, so the regressors are adaptive and temporally dependent rather than independent, and the lagged population state entering the regressor is itself estimated. Both features sit outside the textbook independent-design calculation, and both are exactly the features that matter: a persistent trajectory supplies little directional excitation, and estimation error enters the response and the covariate at once.

Theorem 11 (Covariance-aware rolling-regression uncertainty)**.** *Form the reference-difference design and response matrices from m previous pairs along the trajectory, and let $\widehat{D}_t$ be the resulting pseudoinverse estimate. Suppose the response map is twice continuously differentiable with bounded second derivative, that the true state Jacobian has norm at most $q_0 < 1$, and that the observed design has full row rank. Let Ξ_t collect the distinct population-estimation errors, and let $B_{Y,t}$ and $B_{Z,t}$ be the incidence matrices reproducing the response and lagged-state difference blocks. Writing*

$G_t(q_0) := \left\|B_{Y,t}\widehat{X}_t^\dagger\right\| + q_0\left\|B_{Z,t}\widehat{X}_t^\dagger\right\|$ *and* $\kappa_t := \sigma_{\min}(\widehat{X}_t)$*, then on any event where the sampling errors are bounded by* ε_μ *and* $\|\Xi_t\| \le a_t$*,*

$$\left\|\widehat{D}_t - D_t\right\| \;\le\; \frac{R_t(\varepsilon_\mu)}{\kappa_t} + a_t\, G_t(q_0), \tag{11}$$

where R_t *is an observable curvature allowance. The bound holds pathwise: the design may depend arbitrarily on earlier sampling errors, and no independence between the realized pseudoinverse and* Ξ_t *is required.*

Proof. Expanding about the current point and collecting columns gives $\widehat{Y}_t - D_t\widehat{X}_t = R_t + \Xi_t B_{Y,t} - D_{2,t}\Xi_t B_{Z,t}$. Right-multiply by the pseudoinverse. Retaining the same base matrix Ξ_t in both noise terms, rather than bounding the m differences separately, yields $\left\|\Xi_t B_{Y,t}\widehat{X}_t^\dagger - D_{2,t}\Xi_t B_{Z,t}\widehat{X}_t^\dagger\right\| \le a_t G_t(q_0)$. The argument is pointwise for every realization on the stated event, so no conditioning assumption is needed. □

The incidence representation is what a coordinatewise treatment throws away. The same reference observation appears in every column of the design, so the vectorized covariance of the differences carries an all-ones correction rather than an identity. Carrying that geometry through the observed pseudoinverse, rather than treating correlated differences as unrelated worst-case columns, reduces the median certified radius from 118.5 to 25.6 in the slow and noisy regime, and from 159.9 to 34.9 in the very slow and noisy regime, a factor of roughly 4.6 in both. The resulting certified selector recovers the best evaluated depth in three of four regimes and errs conservatively in the fourth. That fourth case is retained as a negative result, because a bound on sensitivity error is not the same object as a bound on final optimization loss, and the gap between them is instructive.

Joint stability. For the simultaneous update in which the environment and the learner both move, the Jacobian at a fixed point is a block matrix whose off-diagonal entries carry the feedback between them, and stability depends on its spectral radius. Because the depth-K direction enters both off-diagonal blocks, changing the horizon changes the full stability boundary and not merely the accuracy of a direction, so an analysis based on the learner's Hessian alone misreads the online edge of stability. In a scalar coupled model with a quintic learning direction this is fully explicit, with a critical learning rate at which a multiplier crosses -1 and a computable period-two branch beyond it.

6 Experiments

All comparisons are matched-budget. Competing methods receive the same rolling observations, the same number of deployments, the same exploration perturbations, and common random numbers within each seed. Hyperparameters are selected on validation seeds from a shared grid and evaluated on disjoint test seeds.

6.1 MNIST: coverage is not diversity

This is the central empirical result and the one with the clearest operational consequence.

Setup. The task uses MNIST digits 1 and 7 at native 28×28 resolution, giving 13,007 training and 2,163 test images. A logistic classifier trained only on real images attains 99.49% held-out accuracy and serves solely as a class-balance diagnostic. The generator is a compact convolutional

Method at $\beta = .92$	PCA-SWD ↓	balance error ↓	variance ratio
direct (K=0)	$\mathbf{.5067 \pm .0479}$	$.0927 \pm .0439$	$.670 \pm .118$
$K = 1$	$.5121 \pm .0375$	$.0684 \pm .0434$	$.605 \pm .109$
$K = 2$	$.5370 \pm .0451$	$.0507 \pm .0456$	$.552 \pm .101$
$K = 5$	$.6366 \pm .0786$	$\mathbf{.0350 \pm .0292}$	$.445 \pm .082$
full response	$.8845 \pm .1255$	$.0537 \pm .0170$	$.275 \pm .063$

Table 2: Recursive MNIST 1-vs-7 convolutional VAE at $\beta = .92$, mean ± standard deviation over five common seeds. Response depth improves class balance while progressively compressing image variance. The variance column is a diagnostic, not an objective: lower is worse.

variational autoencoder with two strided convolutional encoder layers, a four-dimensional latent variable, and two transposed-convolution decoder layers. Each seed is first trained on real MNIST for 1000 Adam updates at batch size 64, learning rate 10^{-3}, and KL weight 2. The recursive stage then maintains a persistent synthetic panel of 48 decoder outputs that relaxes at rate $\delta = .12$, running for 120 further updates at learning rate 10^{-4}. Within each seed, every horizon is cloned from the same pretrained model and shares data and latent-panel seeds, so the comparison isolates the horizon. Three quantities are reported: a sliced Wasserstein distance computed in a twelve-component principal subspace fitted on real training images, the class-balance error, and the ratio of generated to real average per-pixel variance.

Result. The variance ordering is clean and monotone at every synthetic fraction tested. Reading across horizons from direct training to full response, the ratios are $.693, .651, .615, .541, .413$ at $\beta = .60$; $.679, .623, .577, .482, .327$ at $\beta = .80$; and $.670, .605, .552, .445, .275$ at $\beta = .92$. Deeper response contracts generated variance throughout the range, which is what Proposition 4 predicts.

Class balance moves the other way over most of the range. At $\beta = .92$ the balance error improves from .0927 under direct training to .0350 at $K = 5$, before worsening slightly at full response. Distributional fidelity is not monotone either: at $\beta = .60$ a horizon of one slightly improves the mean sliced Wasserstein distance, from .501 to .497; at $\beta = .80$ it ties direct training; and at $\beta = .92$ direct training is best while full response is substantially worse.

Why this matters. The two diagnostics move in opposite directions over most of the range, and that is the finding. Suppose an operator monitored this system by asking whether both classes were still represented, which is the natural check and a common one. The answer would be healthy, and improving, as the horizon deepened. Meanwhile the model was converging on a single prototype per class. Counting the categories a model can produce is not a measure of how much variety it produces within them.

This also changes what the word collapse denotes here. Full response does not eliminate a class: the balance error remains moderate while generated variance contracts to roughly a quarter of real. The failure is internal to each mode rather than a loss of modes. Any monitoring scheme for a self-consuming pipeline must therefore measure within-mode diversity directly rather than inferring it from coverage.

Scope. Five seeds and two classes. This is a stress test and not a competitive generative benchmark; the paired comparisons are exploratory and are included in the reproducibility package rather than used for significance claims. It establishes that the two failure modes exist and can move independently of one another. It does not characterize the relationship between them, a gap

Regime	Stateful PerfGD	K=1	K=2	K=5
fast, clean (s=16)	$\mathbf{1.0 \times 10^{-5}}$	4.930	2.880	0.341
slow, clean (s=128)	$\mathbf{2.4 \times 10^{-5}}$	5.259	5.200	5.036
slow, noisy (s=128)	61.98 ± 15.43	$\mathbf{6.06 \pm 0.40}$	7.34 ± 0.61	44.10 ± 7.51
very slow, noisy (s=512)	66.97 ± 14.34	6.21 ± 0.55	$\mathbf{5.16 \pm 0.22}$	31.52 ± 10.20

Table 3: Matched-budget comparison on `ToyLinear`. Mean final long-run objective gap; noisy rows report mean ± standard error over 20 fresh test seeds. The exact stationary gain is 4.00, 28.30, and 111.68 at $s = 16, 128, 512$ respectively.

discussed in Section 7.

6.2 Supporting generative experiments

Finite-sample mixtures. Replacing expected mode frequencies by binomial frequencies drawn from batches of 256, over 40 seeds at each synthetic fraction, preserves the collapse ordering predicted by Theorem 2: full response reaches collapse probability one first, followed by $K = 5$ and then $K = 2$. Training all five parameters of the identified mixture reproduces both the weight ordering and the variance-compression ordering of Proposition 4.

State-dependent curation. On the family (8), the final real-target error at $\beta = .95$ is $.1920 \pm .0476$ for direct training, $.1474 \pm .0518$ at $K = 1$, $.1112 \pm .0563$ at $K = 2$, $.1019 \pm .0658$ at $K = 5$, $.2915 \pm .0564$ at $K = 10$, and $.4164 \pm .0441$ at full response. Intermediate horizons beat both endpoints, and the correction-angle measurements confirm that depth is rotating the correction rather than rescaling it.

A two-dimensional autoencoder. On a two-mode target, shallow horizons at $\beta = .8$ reduce the sliced Wasserstein distance from .740 to roughly .658, while full response raises it to .972 and cuts generated variance to 27% of real. Reducing the step size with matched optimization time does not remove the concentration, so it is not a step-size instability.

A retained negative result. A denoising-score pilot with state-dependent curation shows no mean-fidelity benefit from positive response at $\beta = .95$ over eight seeds, with $.336 \pm .112$ for direct training against $.353 \pm .088$ at $K = 5$, although shallow horizons do modestly improve mode balance and reduce run-to-run variability. Finite depth is not automatically beneficial, and this result is reported alongside the positive ones rather than relegated to an appendix.

6.3 Supervised stateful benchmarks

The rolling finite-difference estimator of Stateful PerfGD is reproduced on its released `ToyLinear` environment, in which the population mean relaxes toward a linear target at rate δ, with dimension five, a rolling horizon of fifteen response pairs, and 240 post-warmup deployments. The response time s is defined so that a frozen model leaves one percent of the residual after s steps. The two methods differ in one place only: whether the estimated short-run derivatives are combined through the full stationary inverse or through a truncated sum.

The reversal is sharp. With clean estimates the full stationary sensitivity is essentially exact and dominates every truncation by orders of magnitude. With slow and noisy estimates the ranking inverts, and a horizon of one or two outperforms the full inverse roughly tenfold. The mechanism is

the one Proposition 7 describes: on the same test seeds, the mean error in the estimated stationary sensitivity is 12.76 for Stateful PerfGD against 5.17, 8.34, and 4.80 for horizons of one, two, and five. A nonlinear environment, in which the response depends on the current state through a bounded nonlinearity, reproduces the reversal, which rules out the simplest explanation that the effect is an artifact of affine response.

Truncation is not the only remedy. Explicit Tikhonov and spectral regularization of the stationary inverse recover much of the robustness lost by the raw inverse, and in one slow noisy regime stationary regularization is the best of the methods tested. Finite response is therefore one regularization mechanism among several rather than a unique fix. What distinguishes it is that it is integer-valued and requires no penalty to tune, that it carries an interpretation as a foresight horizon, and that it modifies the joint learner–environment Jacobian rather than only the estimated sensitivity.

7 Discussion and limitations

The conclusion supported here is narrower than a claim of general superiority. When the stationary sensitivity can be estimated accurately, full response is preferable by orders of magnitude, and the clean-estimate experiments show this plainly. Truncation is one regularizer among several. The analytic transition results are exact within a deliberately structured model, and Proposition 6 makes precise that those closed forms are available exactly because the model sits in the non-generic eigenspace-aligned class; Section 4 is what carries the claim outside it. The identified-component mixture is not an unlabeled theorem. The neural evidence is modest in scale and includes a negative control.

The MNIST result opens a question it cannot answer. Two failure modes appear in that experiment, loss of balance and loss of within-mode variety, and they move independently under a sweep over the horizon. How they relate remains unknown. They may be two regions of a single attracting set with transitions between them, in which case a converged system could move from one to the other with no change in its inputs, and monitoring the current state would not be sufficient. Or they may be distinct destinations selected by the horizon setting, in which case the outcome is determined and can be tuned. A direct test is available and inexpensive: run to convergence, hold every input fixed, continue running, and watch for spontaneous movement between them. It has not been run here.

Scaling. The generative experiments use five seeds and two classes, with pixel variance standing in for diversity. A CIFAR-10 diffusion study sweeping the horizon and the synthetic fraction, reported with precision and recall rather than a single fidelity score, is the natural next step, since those metrics separate fidelity from diversity directly. The prerequisite is a way to propagate state sensitivity through a multi-step denoising chain, which naive backpropagation makes memory-prohibitive; the pilot above sidesteps this by tracking summary coordinates rather than the full panel. That prerequisite is a methodological question of independent interest.

Reinforcement learning. In on-policy reinforcement learning the data distribution is endogenous by construction. The policy gradient theorem contains a term for the dependence of the state distribution on the policy parameters, and standard implementations omit it (Nota and Thomas, 2020). That omission is structurally the zero-horizon endpoint of the present framework, and whether the exact object is desirable in that setting appears not to have been examined.

8 Conclusion

The stationary response is not always the right target for a learner whose deployment changes its own data. Two distinct reasons have been given.

The exact hypergradient can be undesirable. In a self-consuming generative loop, fully internalizing one's own feedback is exactly equivalent to optimizing an objective that rewards reducing the entropy of one's own output. The symmetry-breaking threshold falls from a synthetic fraction of one at zero foresight to one half at full foresight, with a matching closed-form law for how much within-mode variance is lost along the way. Nothing is approximated and no quantity is noisily estimated: the optimizer is correct and the objective it implies is not. This is a specification failure, and better data does not remedy it.

The exact hypergradient can be unestimable. Its gain is a resolvent norm that diverges as the environment slows, so the method is least reliable exactly where it is most needed. A finite horizon trades a known geometric bias for bounded variance, and under matched budgets the ranking between the two reverses between clean and slow-noisy estimation. This is an estimation failure, and better data does remedy it.

Monitoring must measure within-mode diversity directly. The MNIST stress test shows response depth improving class balance while compressing generated variance to 27.5% of real. Both classes survive; each collapses internally. A monitoring scheme based on mode coverage would certify a healthy system throughout. This is the most immediately actionable finding here for anyone operating a self-consuming training pipeline today.

Taken together, these results suggest that the response horizon is best treated as a modeling choice about how much of its own influence a learner should internalize, rather than as an approximation parameter to be driven to its limit. That choice has consequences for the objective being optimized and for the stability of the coupled dynamics alike.

References

Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári. Online least squares estimation with self-normalized processes. *arXiv:1102.2670*, 2011.

S. Alemohammad et al. Self-consuming generative models go MAD. In *ICLR*, 2024.

M. Blondel et al. Efficient and modular implicit differentiation. In *NeurIPS*, volume 35, 2022.

G. Brown, S. Hod, and I. Kalemaj. Performative prediction in a stateful world. In *AISTATS*, volume 151, pages 6045–6061, 2022.

S. Cai, F. Han, and X. Cao. Performative control for linear dynamical systems. In *NeurIPS*, volume 37, 2024.

Z. Cai et al. Stabilizing self-consuming diffusion models with latent space filtering. In *AAAI*, 2026.

E. Dohmatob, Y. Feng, and J. Kempe. Strong model collapse. In *ICLR*, 2025.

S. Fu et al. Towards theoretical understandings of self-consuming generative models. In *ICML*, volume 235, pages 14228–14255, 2024.

M. Gerstgrasser et al. Is model collapse inevitable? Breaking the curse of recursion by accumulating real and synthetic data. *arXiv:2404.01413*, 2024.

N. Gillman et al. Self-correcting self-consuming loops for generative model training. In *ICML*, volume 235, pages 15646–15677, 2024.

R. Grazzi, L. Franceschi, M. Pontil, and S. Salzo. On the iteration complexity of hypergradient computation. In *ICML*, volume 119, pages 3748–3758, 2020.

R. Grazzi, M. Pontil, and S. Salzo. Nonsmooth implicit differentiation: deterministic and stochastic convergence rates. In *ICML*, volume 235, pages 16250–16274, 2024.

Z. Izzo, L. Ying, and J. Zou. How to learn when data reacts to your model: performative gradient descent. In *ICML*, volume 139, pages 4641–4650, 2021.

Z. Izzo, J. Zou, and L. Ying. How to learn when data gradually reacts to your model. In *AISTATS*, volume 151, pages 3998–4035, 2022.

J. Kazdan et al. Collapse or thrive? Perils and promises of synthetic data in a self-generating world. In *ICML*, volume 267, pages 29469–29494, 2025.

N. B. Khelifa, R. E. Turner, and R. Venkataramanan. Recursively trained diffusion models: limiting collapse distribution and spectral characterization. *arXiv:2606.13796*, 2026.

L. Lin and T. Zrnic. Plug-in performative optimization. In *ICML*, volume 235, pages 30546–30565, 2024.

H. Liu, Q. Li, and H. T. Wai. Two-timescale derivative free optimization for performative prediction with Markovian data. In *ICML*, volume 235, pages 31425–31450, 2024.

L. Metz et al. Understanding and correcting pathologies in the training of learned optimizers. In *ICML*, volume 97, pages 4556–4565, 2019.

M. Mofakhami, I. Mitliagkas, and G. Gidel. Performative prediction with neural networks. In *AISTATS*, volume 206, pages 11079–11093, 2023.

C. Nota and P. S. Thomas. Is the policy gradient a gradient? In *AAMAS*, pages 939–947, 2020.

J. C. Perdomo, T. Zrnic, C. Mendler-Dünner, and M. Hardt. Performative prediction. In *ICML*, volume 119, pages 7599–7609, 2020.

G. Piliouras and F.-Y. Yu. Multi-agent performative prediction: from global stability and optimality to chaos. In *ACM EC*, pages 1047–1074, 2023.

M. Rudelson and S. Zhou. Errors-in-variables models with dependent measurements. *Electronic Journal of Statistics*, 11(1):1699–1797, 2017.

A. Shaban, C.-A. Cheng, N. Hatch, and B. Boots. Truncated back-propagation for bilevel optimization. In *AISTATS*, volume 89, pages 1723–1732, 2019.

I. Shumailov et al. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759, 2024.

M. Simchowitz et al. Learning without mixing: towards a sharp analysis of linear system identification. In *COLT*, volume 75, pages 439–473, 2018.

A. T. Suresh, A. Thangaraj, and A. N. K. Khandavally. Rate of model collapse in recursive training. In *AISTATS*, volume 258, pages 1396–1404, 2025.

Y. Wang, Z. Xu, T. Zhao, and M. Tao. Good regularity creates large learning rate implicit biases: edge of stability, balancing, and catapult. *JMLR*, 26(273):1–68, 2025.

X. Wei and X. Zhang. Self-consuming generative models with adversarially curated data. In *ICML*, volume 267, pages 66300–66321, 2025.